SINA SADRIAN

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SKILLS

- **Analog/Digital/Power Electronics Design:** schematic & layout design (Altium Designer, Cadence), simulation (SPICE, PSIM), component selection, DFM, DFT, system-level & board-level troubleshooting
- **Regulatory Compliance:** EMC emissions (conducted, radiated), immunity (ESD, surge, EFT/B), UL safety
- **Communication Interfaces (hardware & firmware):** Ethernet, USB, SPI, CAN, UART, RS-485, I2C
- **Test & Task Automation:** Python (PyVISA, SCPI for controlling lab equipment, PyQt for GUI development)
- **Programming**: C/C++, Python, Git

EXPERIENCE

Electrical Engineer II April 2025 - Present
Amazon - Physical Stores Tech. *Seattle, WA*

- Electrical Engineering lead for an automated entry/exit gate development program for cashierless retail stores, driving development from concept to pilot readiness across three PCBs covering power and sensor interfaces
- Defined the overall electrical architecture for the gate system, including board partitioning, power conversion, supercapacitor-based backup power, cable harnessing, and integration of Android compute modules, sensors, motor subsystems, and power subsystems over various communication channels (USB, I2C, RS-485, Ethernet)
- Owned full PCB development, including circuit design, component selection, schematic capture and PCB layout (Altium Designer), prototyping, board bring-up, and troubleshooting
- Worked directly with cross-functional teams on product development and with contract manufacturers and vendors to support manufacturing builds

Senior Electronics Engineer April 2022 - April 2025
Motorola Solutions Inc. *Vancouver, BC*

- Electronics hardware design for developing a wide range of medium to high-volume IP video security cameras
- Was responsible for the power-supply board PCBs for multiple projects, accommodating switching power supply converters (flyback, buck, ranging from 10W to 80W depending on the project), input front-end (PoE and AC/DC), and communication channels (including Gigabit Ethernet)
- Each project involved various stages from product concept through to volume production, including high-level architecture design, schematic & PCB layout design (Altium Designer), circuit simulation (mostly LTspice), component selection, prototyping, validation testing & debugging, SMT component soldering & reworking, documentation & version control (SVN, Agile PLM), regulatory compliance approvals, manufacturing support
- Troubleshooted multiple problems and proposed solutions to pass regulatory compliance approvals, including EMC emissions (conducted, radiated), immunity (ESD, surge, EFT/B), UL safety (hi-pot, critical components)
- Initiated and developed the automation of the numerous test protocols used for power-supply board design evaluations, resulting in significant savings in PCB design timelines and allocated resources. Development was done using Python (PyVISA, SCPI for controlling lab equipment such as power supplies and oscilloscopes, PyQt for GUI development, NumPy for analysis, XlsxWriter)
- Position was ”Electrical Designer” from 2022 to 2024; was promoted to ”Senior Electronics Engineer” since 2024

Electronics Hardware Engineer Nov. 2020 - April 2022
Rigid Robotics Inc. *Burnaby, BC*

- Developing precise positioning solutions for the mining industry, using UHF radio & GNSS satellite signals
- Responsible for electronics hardware and firmware design and troubleshooting
- PCB boards were essentially responsible for interfacing sensors such as GNSS/GPS and IMU with the processors and providing various communication channels (such as Ethernet, WIFI, USB, SPI, CAN, UART, RS-485, I2C)
- Electronics hardware responsibilities included design, design review (Altium Designer), testing & troubleshooting of PCB boards. Helped resolve multiple issues in the earlier versions of PCB boards.
- On the embedded firmware side, worked on refining the current code base as well as adding new features and firmware building blocks. Programming was in C/C++ and Python. Target processors included both a microcontroller (Microchip SAM-E70 family) and an embedded Linux processor (TI AM3358). Version control was done with Git.

Research Assistant, PhD Student May 2020 - April 2021
University of British Columbia, Power Electronics Lab. *Vancouver, BC*

- PhD researcher at the "Power Electronics" laboratory (Dr. Martin Ordonez)
- University-Industry collaboration focused on high-power battery charging for electric vehicles

Electrical Engineer Aug. 2019 - April 2020
Advanced Intelligent Systems Inc. *Burnaby, BC*

- Member of the electrical team, developing autonomous robots for the horticulture industry
- Responsibilities included embedded programming, circuit and PCB design, troubleshooting
- Worked on multiple modules such as stepper motor and brushless DC motor drivers, ultrasonic sensors, IMU, power supply, cabling and interconnects
- PCB design with Altium Designer

Research Assistant, MASc Student Sept. 2016 - Aug. 2019
Simon Fraser University, Power Electronics Lab. *Surrey, BC*

- Proposed a novel solution to improve the power-density and efficiency of AC/DC converters
- MASc thesis project; can be accessed at bit.ly/sina-thesis.
- Results also published in the peer-reviewed IEEE OJPEL journal. Link: bit.ly/sina-ieee
- Developed a 400V/800W prototype circuit to demonstrate the operation of my proposed solution
- Circuit design and simulation with PSIM and MATLAB/SIMULINK
- PCB design with Altium Designer

EDUCATION

M.A.Sc. in Mechatronic Systems Engineering Sept. 2016 - Apr. 2019
Simon Fraser University *Surrey, BC*

- Supervisor: Dr. Jiacheng (Jason) Wang
- Recipient of the "Graduate Fellowship" award

B.Sc. in Electrical Engineering Sept. 2011 - Feb. 2016
Sharif University of Technology *Tehran, Iran*